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Learning Beyond Still Frames: Scaling Vision-Language Models with Video

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Abstract

High-quality image-text data is critical for Vision-Language Models (VLMs), yet traditional image-based pretraining is resource-intensive and fails to capture the temporal dynamics needed for video understanding. To address this, we introduce video pretraining to enhance VLMs with temporal reasoning. We propose Causal Hierarchical Aggregation, a novel method that efficiently processes video by separating computationally heavy spatial encoding from lightweight temporal propagation. This technique builds hierarchical receptive fields, enabling effective learning from large-scale video data. Scaling our method to over 100 billion video tokens, we achieve state-of-the-art performance and high throughput on both image and video understanding tasks (Figure 1). Our approach offers a scalable solution to advance multimodal learning for dynamic contexts. Our code and pretrained models will be released at https://github.com/invictus717/LLaVA-Prime.

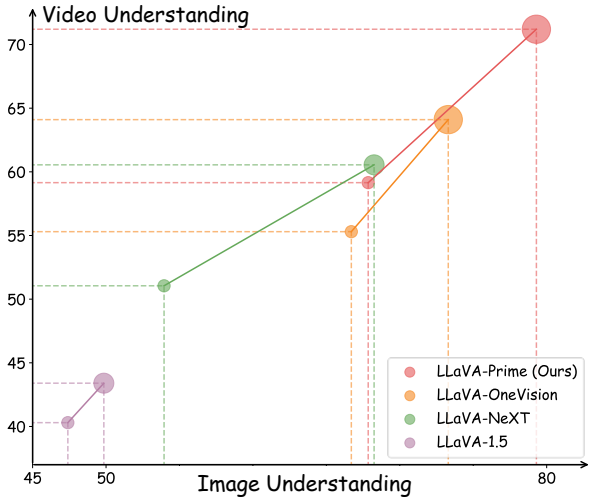


Figure 1. **A Comparison between LLaVA-Prime and existing VLM methods**, where the circles in different sizes denote 7B, 13B, and 72B language models, respectively. We propose to introduce scalable video pretraining for the vision encoder and large language model, effectively improving image and video understanding performance. We choose the MMVet, MMBench, Egoschema, and VideoMME benchmarks as representatives.

1. Introduction

The advancement of vision-language models (VLMs) significantly enhanced the capabilities of the large language models (LLM) in multimodal understanding [2, 7, 26, 44, 46, 56, 61, 66]. Existing VLMs, including LLaVA [44, 46], Gemini [61], and Qwen-VL [2], primarily rely on cross-modal alignment between vision encoders and language models. As image-text datasets used for pretraining scale-up, VLMs demonstrate impressive scalability in universal vision-language understanding, with significant advancements in visual question answering [66], visual knowledge learning [44], and visual agents [2, 65].

The effectiveness of this development approach is *intrinsically tied to the quality of the pretraining data*, specifically the image-text pairs, which must feature both high visual fidelity and accurate, contextually relevant captions. However, current pretraining datasets face two critical limitations. **First**, creating high-quality image-text pairs with coherent natural language captions is an extensive endeavor, demanding significant financial, computational, and storage resources. Each aspect of data quality refinement adds both cost and complexity, making high-quality datasets a challenging and often impractical standard for widespread use. **Second**, and more fundamentally, we observe that while these high-quality datasets enhance static image understanding, they fall short in supporting dynamic video comprehension within vision-language models. Image-text pairs, by nature, *capture isolated moments without temporal context*, which limits the models' ability to perceive sequential changes essential for video understanding.

In other words, static vision-language data lacks the temporal depth needed to understand motion and evolving scenes, limiting VLMs in tasks that require a mul-

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timodal understanding of sequential dynamics, such as event progression, object interaction, and causality within videos. Meanwhile, video analysis demands complex spatial-temporal modeling on both computation burdens and strict temporal causality. These limitations highlight the need for VLMs to integrate temporal dynamics, enabling models to not only perceive what is present in a scene but also understand how it evolves over timestamps by learning from video data.

To address the challenges, we aim to propose a scalable image-video joint pretraining framework to enhance both spatial and temporal understanding, allowing models to learn richer, context-aware representations, which is the approach that remains largely under-explored in the community. This approach has two core advantages: **1**) *Enhanced Visual Cognition*: Videos inherently provide a richer level of visual information than static images. Each video consists of a sequence of temporally connected frames, enabling models to capture not only individual visual snapshots but also the flow and evolution of scenes over time. Pretraining with video data, therefore, fosters a more comprehensive and nuanced understanding of visual contexts, enhancing general visual cognition. **2**) *Improved Multimodal Alignment*: Videos inherently demand more descriptive and detailed captions than images, as they depict complex, unfolding events. This demand for longer, more contextually rich textual descriptions requires more fine-grained and temporal alignment between vision and language. By aligning with temporally detailed captions, VLMs trained on video data can achieve a deeper integration of visual and linguistic information, resulting in improved vision-language alignment essential for complex multimodal tasks. Incorporating video pretraining into the VLM pipeline not only advances general visual understanding but also addresses the temporal limitations inherent in static image-text pairs, paving the way for VLMs with more robust multimodal capabilities.

Therefore, we propose LLaVA-Prime, an innovative extension of the LLaVA [44, 45] architecture that incorporates video data into pretraining. There are four stages in this framework: 1) video pretraining with a vision encoder to capture temporal visual dynamics; 2) video pretraining with a large language model to integrate sequential visual information with linguistic context; 3) joint image-video modeling with curated image-video-text pairs to enhance synchronization between visual and textual modalities; 4) visual instruction tuning. Specifically, we propose the Causal Hierarchical Aggregation (CHA) to achieve outstanding computation efficiency and temporal causality. In CHP, videos can be encoded using the same image encoder, with only a lightweight temporal block added to efficiently aggregate redundant video tokens. This approach ensures that the number of video tokens remains consistent with standard image tokens while strictly preserving 1D temporal causality.

We conduct extensive experiments on video-pretrained vision encoders, video-pretrained LLMs, and fully integrated video-pretrained VLMs. On the CLIP benchmarks, the video-pretrained vision encoder demonstrates impressive zero-shot visual recognition capabilities, achieving competitive performance with larger models: *with only 1B parameters, it outperforms the InternVL-ViT-6B model [7] and rivals the EVA-CLIP-8B model [64]*. Following vision-language pretraining and instruction tuning, *LLaVA-Prime achieves new state-of-the-art performance on leading vision-language benchmarks*, as shown in Figure 1, including MME [21], MMBench [47], and MMVet [77], among others, as well as on video understanding benchmarks such as EgoSchema [51], VideoMME [22], ActivityNet [78], and other benchmarks.

2. Related Work

Vision-Langauge Models. In the recent two years, the emergence of large language models [55, 56, 61, 66] has significantly improved the reasoning and understanding abilities of machine intelligence. Pioneers such as Qwen-VL [2] and LLaVA [46] have shown promising potential in the visual instruction tuning paradigm. As image-text data are scaled, VLMs present impressive vision-language understanding capacity. However, such development relies on pretraining data quality, which makes it costly and difficult to scale up further. To address this limitation, we aim to propose a new pretraining paradigm with other pretraining data (video), that is more data-efficient, scalable, and beneficial to VLMs.

Video Pretraining. VideoCLIP [74] initially demonstrates the potential of zero-shot video recognition through pretraining on video data, paving the way for advancements in joint video-image training. OFA [70] and subsequent works [69, 71, 79, 80] have further extended these capabilities, showing that video pretraining can yield significant benefits for video-language understanding tasks. Recent models like VideoChat [41] and LLamaVid [42] have successfully integrated video dynamics into VLMs, enabling some degree of temporal awareness in understanding visual sequences. However, these efforts fall short of capturing the full potential of temporal context pretraining — the ability to understand complex, evolving temporal patterns and dependencies across frames, which is critical for nuanced comprehension of real-world events in video. Achieving this remains challenging due to several key factors: **1**) the data-efficient video-pertaining method, how to efficiently learn temporally coherent sequences; **2**) the limitations of current pretraining which are often optimized for static image or short sequence processing and struggle to generalize to longer temporal contexts. As a result, while video pre-

training has seen promising advances, truly robust temporal context understanding in VLMs remains an open challenge.

In this paper, we delve into the *critical role* of video data in VLM pretraining, examining why video data is essential for advancing multimodal understanding. We explore the optimal data scale required for effective video pretraining, assess the *specific impact of video pretraining on both vision encoders and LLMs*, and highlight the significant improvements video pretraining brings to VLM models. Through this analysis, we aim to provide a comprehensive understanding of the promising potential of video data in enhancing VLM performance.

3. Method

We propose LLaVA-Prime, a scalable framework to enable LLaVA with video data pretraining, which leads to state-of-the-art image and video understanding abilities simultaneously. Starting from the LLaVA baseline, which includes a vision encoder pretrained by CLIP [59] and an LLM pretrained with Qwen2 [2], our approach leverages video-text pairs to enhance multimodal learning. The pretraining process unfolds in 4 stages: first, pretraining the vision encoder with video (Stage I); second, pretraining the LLM on video-text pairs (Stage II); and finally, training the VLM with the causal hierarchical aggregation (Stages III & IV).

Stage	Model	Data	Type	Loss	Settings
Stage I	ViT-g	Panda-70M	Video	InfoNCE	Finetune
Stage II	LLM-7B	Panda-70M	Video	Next-token	Finetune
Stage III	ViT+LLM	LLaVA-1.5	Image-Video	Next-token	Finetune+CHA
Stage IV	ViT+LLM	LLaVA-1.5	Image	Instruction	Freeze ViT

Table 1. Overview of LLaVA-Prime pretraining configurations.

Baseline Settings. We start from the LLaVA-1.5 baseline, replace a large vision encoder with 1B parameters for stronger capacity with video and image, and use Qwen2-7B [2] as our LLM. We summarize the training data, model, and parameter scales in Table 1.

3.1. Stage I: Pretraining Video Encoder

Given the video dataset $\mathcal{D}_V = \{(\boldsymbol{x}_v, \boldsymbol{x}_t)\}^N$ with video $\boldsymbol{x}_v$ and paired caption $\boldsymbol{x}_t$, we first pretrain the CLIP vision encoder $f_{vit}(\boldsymbol{x}_t; \theta_{vit})$ using contrastive learning [59], where we use the CLIP text encoder as the text branch for the InfoNCE [59] loss function. For each video, we select only 4 key frames along with the caption, representing the frames as $\boldsymbol{x}'_v \in \mathbb{R}^{4\times3\times H\times W}$. We then reshape $\boldsymbol{x}'_v$ into a temporally relevant sequence of images, resulting in $\boldsymbol{x}'_v \in \mathbb{R}^{12\times H\times W}$. Next, we feed $\boldsymbol{x}'_v$ and $\boldsymbol{x}_t$ to the vision encoder and text encoder, respectively, obtaining the video vector $\boldsymbol{z}_v$ and text vector $\boldsymbol{z}_t$. We fine-tune the vision encoder $f_{vit}(\boldsymbol{x}_t; \theta_{vit})$ as follows:

$$\mathcal{L}_{vit} = -\frac{1}{N_B}\sum_{i=1}^{N}\log\frac{\exp(\text{sim}(\boldsymbol{z}_v^{(i)}, \boldsymbol{z}t^{(i)})/\tau)}{\sum_{j=1}^{N}\exp(\text{sim}(\boldsymbol{z}_v^{(i)}, \boldsymbol{z}_t^{(j)})/\tau)}, \quad (1)$$

where N_B is the batch size, $\text{sim}(\cdot,\cdot)$ is the cosine similarity, and τ is a temperature hyper-parameter.

Model	MMLU (%)	CMMLU (%)	MMLU-Pro (%)
Qwen-2-7B	68.2	49.4	27.6
Qwen-2-7B-V	48.6	52.8 (+3.4)	28.6 (+1.0)

Table 2. **Text Evaluation of LLMs**. We find video pretraining can surprisingly improve performances on certain benchmarks.

3.2. Stage II: Video Pretraining LLM

Stage II is to pretrain the LLM with video data, making it more adaptive to vision tokens. To achieve this, we utilize patch embedding layers for video tokenization and apply a next-token prediction loss to pretrain the LLM with the video-text dataset, as described in § 3.1. After tokenization, we interleave video token and text token embedding sequences $\boldsymbol{s} = (\boldsymbol{e}_v^1, \boldsymbol{e}_v^2, \ldots, \boldsymbol{e}_v^{N_v}, \boldsymbol{e}_t^1, \boldsymbol{e}_t^2, \ldots, \boldsymbol{e}_t^L)$, where N_v and L are the numbers of vision tokens and sentence length, aiming to predict each next token using the next-token prediction loss:

$$\mathcal{L}_{\text{NTP}} = -\sum_{i=1}^{N_v+L-1}\log p_\theta(s_{i+1}|s_1, s_2, \ldots, s_i), \quad (2)$$

where $p_\theta(s_{i+1}|s_1, s_2, \ldots, s_i)$ is the probability assigned by the LLM to the next token given the preceding tokens. This objective encourages the LLM to learn from both vision and text tokens, making it more adaptable to multimodal inputs. To stabilize the pretraining and avoid the spike during this process, we use the patch aligning layer proposed by EVE [14]. And we use the video-pretrained vision encoder in Stage I (§ 3.1) to distill the LLM weights on vision tokens, which is a token-wise mean squared error:

$$\begin{aligned} \boldsymbol{z}_v &= f_{vit}(\boldsymbol{x}_v, \theta), \quad \boldsymbol{e}_v = f_{llm}(\boldsymbol{x}_v, \theta'), \\ \mathcal{L}_{distill} &= \frac{1}{N_v}\sum_{i=1}^{N_v}\|\boldsymbol{z}_{v,i} - \boldsymbol{e}_{v,i}\|_2^2. \end{aligned} \quad (3)$$

Surprisingly, we find video pretraining for LLM can improve text capacity on the MMLU [28], CMMLU [39], and MMLU-Pro [72] benchmarks, as shown in Table 2, which results in gains of +3.4% and +1.0% for CMMLU and MMLU-Pro, respectively.

3.3. Scalable Joint Image-Video Training

Architecture for Image and Video Encoding. Generally, we use a vision encoder to encode images; we input an image $x_i \in \mathbb{R}^{3\times H\times W}$ and it outputs image tokens

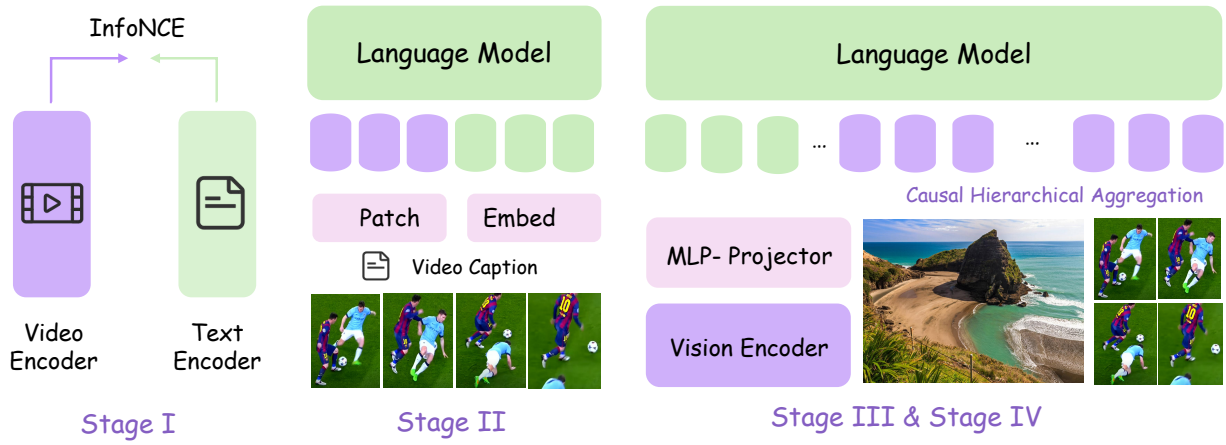


Figure 2. **Framework Overview of LLaVA-Prime**. We introduce video pretraining for the VLM framework in Stage I and Stage II. Then, we perform joint image-video-text pretraining and instruction tuning.

$z_i \in \mathbb{R}^{1,N_v,D}$, where D is the embedding dimension. For a video, we use the same vision encoder to encode videos; we input an video $x_v \in \mathbb{R}^{T\times 3\times H\times W}$ and it outputs video tokens $z_v \in \mathbb{R}^{1\times(N_v\cdot T)\times D}$, where T means the time dimension. After that, we use the causal hierarchical aggregation module to compress video tokens to $z_v \in \mathbb{R}^{1\times N_v\times D}$.

Causal Hierarchical Aggregation. The first step of causal hierarchical aggregation is to model the redundant video by temporal 1D causality. We follow the idea of run-length tokenization [9] and model temporally consecutive video as the first frame x_v^0 and corresponding dynamic sequences $[\Delta x_v^{0,1}, \Delta x_v^{1,2}, \cdots, \Delta x_v^{T-1,T}]$, where $\Delta x_v^{i,j} = x_v^j - x_v^i$. Then we set up the causal masks as zero vectors M_i with the total length of T and $M_{ij} = 1$ if $j \leq i$. And we employ a 1D convolution layer to perform the 1D-causal aggregation among the $N_v \times T$ video tokens to obtain the video features with strict causality:

$$\boldsymbol{z}_v = \texttt{CausalConv1D}(x_v^0, \Delta x_v^{0,1}, \Delta x_v^{1,2}, \cdots, \Delta x_v^{T-1,T}), \tag{4}$$

in the $\texttt{CausalConv1D}(\cdot)$ layer, these temporal features are aggregated as:

$$\boldsymbol{z}_v^0 = w \cdot x_v^0, \quad \boldsymbol{z}_v^i = \boldsymbol{z}_v^{i-1} \cdot M_i \cdot w \cdot \Delta x_v^{i-1,i}, \tag{5}$$

Multi-Scale Causal Dilated Convolutions. Indeed, causal temporal modeling can employ various network types; however, utilizing 1D-causal convolutional layers offers enhanced flexibility and computational efficiency through hierarchical design. Specifically, we propose two advancements: **1**) Employing 1D dilated convolutional layers for causal aggregation modeling, significantly enhancing flexible temporal modeling across diverse time ranges. **2**) Recognizing the importance of temporal receptive fields in video learning and inspired by the large-kernel design [15], we integrate large-kernel configurations to enhance the aggregation of video tokens. Therefore, we employ several dilated ($d = [1, 3, 5, 7]$), large-kernel ($k = [13, 13, 9, 7]$), 1D-causal convolutional layers to aggregate video tokens,

Method	MME	MMB	Egoschema	VideoMME
LLaVA-1.5	1510	64.3	38.5	42.1
(a) LLaVA-1.5 (ViT-g+Qwen2-7B)	1569	70.6	45.0	55.1
Only using LLaVA-1.5 data for VLM				
(b) +Scale up video data in Stage I	1576	72.2	49.6	57.3
(c) +Scale up video data in Stage II	1584	72.3	45.9	56.4
(d) +Scale up video frames in Stage I	1593	73.1	48.4	56.8
(e) +Scale up video frames in Stage II	1588	72.8	49.7	57.2
Scaling VLM Data				
(f) +Scale up context length	1587	73.5	48.2	56.9
(g) +Scale up video resolution	1543	70.4	46.1	54.8
(h) +Scale up frames to 1 fps	1555	71.2	44.9	57.5
(i) +Scale up frames to 2 fps	1533	70.8	48.2	58.4
(j) +Scale up frames to 5 fps	1577	73.2	51.2	59.8
Architecture Ablation				
(k) +1D Causal aggregation	1584	72.3	46.8	55.7
(l) +Hierarchical temporal fusion	1579	72.7	47.3	55.9
Framework Ablation				
(m) +Stage I video pretraining	1603	74.3	51.2	57.5
(n) +Stage II video pretraining	1621	73.5	52.4	58.8
(o) +Stage III CHP	1610	73.6	54.3	60.2
Scaling Factors				
(p) +Scaling LMs	1723	81.2	53.5	62.8
(q) +Scaling Image tokens	1643	75.3	50.5	54.7
(r) +Scaling Video tokens	1623	74.2	53.5	58.9

Table 3. **Ablation Study of LLaVA-Prime**. We conduct experiments on both image and video evaluation benchmarks. We first explore scaling up video pretraining data in Stages I & II, (b) and (c) scale video samples 2M to 16M, (d) and (e) scale video frames from 4 to 8. Then, we explore scaling up VLM data on context length (from 8k to 32k in (f)), video resolution (224 to 336 in (g)), and scaling up video frames to 1 fps in (h), 2 fps in (i), and 5 fps in (j). In addition, we ablate architecture design in (k) and (l), our proposed Stage I & Stage II & Stage III in (m), (n), and (o), respectively. Finally, we present the scaling results on LMs in (p), image tokens in (q), and video tokens in (r).

method	ImageNet-1K [13]	ImageNet-V2 [60]	ImageNet-Adv. [30]	ImageNet-Ren. [29]	ImageNet-Ske. [68]	ObjectNet [3]	CIFAR-10 [34]	CIFAR-100 [34]	MNIST [36]	Caltech101 [20]	SUN397 [73]	FGVC Aircraft [50]	Country-211 [59]	Stanford Cars [33]	DTD [10]	EuroSAT [27]	FER2013 [23]	Flowers-102 [54]	Food-101 [4]	GTSRB [63]	PCam [67]	Pets [57]	Rendered SST2 [59]	RESISC45 [8]	STL-10 [11]	VOC2007 [16]	**avg. top-1 acc.**
EVA-01-CLIP-g/14 [18]	78.5	71.5	73.6	92.5	67.6	72.3	98.3	88.7	62.6	87.7	74.2	32.4	28.9	91.7	61.7	73.8	52.2	74.5	93.5	49.3	49.9	94.2	58.4	70.3	98.9	85.7	72.4
EVA-01-CLIP-g/14+ [18]	79.3	72.5	74.1	92.7	68.4	75.3	99.1	90.1	72.0	89.5	74.7	39.9	31.8	90.7	67.8	73.2	56.0	79.7	93.7	66.5	62.4	94.9	58.6	71.4	99.5	84.7	75.3
OpenCLIP-G/14 [32]	80.4	73.6	69.3	92.8	69.9	73.0	98.3	87.5	71.6	89.4	75.0	53.6	34.9	94.9	69.1	71.1	**59.6**	81.5	93.1	62.7	63.6	95.3	65.3	72.6	98.5	87.4	76.3
InternVL-C-6B [6]	83.2	77.3	83.8	**95.7**	74.3	80.6	**99.4**	93.1	80.6	89.5	76.3	53.3	35.1	94.4	70.8	79.4	56.2	85.8	95.3	65.5	48.7	96.3	**68.4**	74.4	99.4	80.0	78.3
DFN5B-CLIP-H/14 [17]	83.5	77.4	71.7	92.9	72.8	76.7	98.8	90.5	**85.8**	89.5	77.0	71.4	34.4	95.8	70.7	65.2	54.7	**92.5**	95.8	67.7	65.2	96.5	54.8	76.1	98.9	81.5	78.3
EVA-02-CLIP-E/14+ [19]	82.1	75.7	82.1	94.7	72.2	79.6	99.3	93.2	74.7	**90.5**	75.3	58.7	37.0	94.7	68.2	75.9	59.0	84.5	94.9	67.7	64.4	96.0	62.6	75.7	99.3	**87.9**	78.7
DFN5B-CLIP-H/14+ [17]	**84.3**	**78.3**	79.6	93.6	73.3	79.6	98.8	90.5	83.6	88.9	77.4	**72.5**	37.9	**96.0**	70.9	61.1	56.1	91.6	**96.2**	67.9	**69.6**	**96.8**	55.5	75.9	99.1	81.9	79.1
EVA-02-CLIP-8B [64]	83.5	77.7	85.2	95.3	74.3	81.2	99.3	92.3	84.8	89.6	76.2	60.5	41.7	94.8	71.0	68.9	56.1	86.4	95.5	70.9	58.1	96.4	66.2	75.3	99.3	85.1	79.4
EVA-02-CLIP-18B [64]	83.8	77.9	**87.3**	**95.7**	**74.7**	**82.2**	**99.4**	**93.8**	83.0	89.8	**77.7**	59.7	**43.1**	94.9	**72.1**	**79.8**	59.3	86.0	95.8	**72.4**	65.2	96.1	67.5	**76.9**	**99.6**	85.8	**80.7**
LLaVA-Prime-ViT-g	82.6	76.4	79.6	94.4	72.1	77.9	99.1	91.8	83.4	86.1	75.6	49.7	36.5	94.7	69.7	77.3	58.7	87.2	94.8	69.5	63.4	95.9	66.9	75.0	99.2	84.2	78.5

Table 4. **zero-shot image classification performance on 27 datasets.** We report top-1 accuracy on all datasets. Impressively, LLaVA-Prime-ViT-g even outperforms InternVL-C-6B and DFN5B-CLIP-H.

which ensures strict temporal causality and fine-grained spatial-temporal granularities.

3.4. Scalability of LLaVA-Prime

We examine the scalability of video pretraining and identify factors that influence its effectiveness in enhancing both image and video understanding of VLMs.

Remark 1: Scaling up video data. As shown in Table 3, we compare a baseline configuration (a) with models that scale up video data in Stage I (b) and Stage II (c). The findings suggest that **1**) both stages of video pretraining are scalable, offering improvements in image and video comprehension simultaneously. **2**) Video data is highly efficient; compared to the vast amounts of text-image data typically required, pretraining with video data yields substantial benefits in model capacity for both image and video understanding.

Remark 2: Scaling up video frames. In comparing settings (h), (i), and (j), we find that the number of video frames is critical to learning temporal dynamics. It's beneficial to pretrain VLMs with longer video frames, which improves the temporal understanding abilities.

Remark 3: Context length. Beyond data scale and quality, we observe that context length in VLMs significantly impacts model performance, especially in video comprehension. Managing context length carefully can improve the model's ability to handle sequential visual information effectively.

4. Experiments

4.1. Setup

We evaluate our models using two distinct benchmark types: 1. Vision Encoder Evaluation: We use the CLIP benchmark, which comprises 26 datasets for zero-shot image recognition. Additionally, we include the Flick30K and MSCOCO datasets to evaluate zero-shot image-text retrieval capabilities. Beyond static image tasks, our model is further tested on zero-shot video recognition using the K400, K600, and K700 datasets, providing a comprehensive measure of our model's efficacy in both image and video recognition domains. 2. Vision-Language Model Evaluation: We conduct a thorough evaluation of our model's performance on a diverse range of benchmarks relevant to vision-language understanding. This includes tasks in visual question answering, visual knowledge comprehension, and video-series temporal context understanding, enabling us to gauge the model's capabilities in complex, multimodal reasoning and temporal analysis tasks.

Method	Frames	K400		K600		K700	
		top-1	AVG	top-1	AVG	top-1	AVG
CLIP [59]	8	58.4	70.1	55.1	67.2	46.1	58.4
EVA-CLIP-L [18]	1	-	65.0	-	64.9	-	59.1
EVA-CLIP-E [18]	1	-	69.8	-	69.3	-	63.4
VideoCoCa-g [75]	16	72.0	81.3	-	-	-	-
InternVL-6B [6]	8	69.1	79.4	68.9	78.8	60.6	71.5
EVA-CLIP-18B [64]	16	-	79.4	-	79.4	-	72.2
VideoPrism-1B [81]	16	**76.4**	**85.4**	-	-	-	-
InternVideo2$_{clip}$-1B	8	73.1	82.4	**72.8**	**81.8**	**64.9**	**75.2**
LLaVA-Prime-ViT-g	8	73.5	82.6	71.3	80.4	62.0	73.5

Table 5. **zero-shot video classification results on Kinetics datasets**. We report the top-1 accuracy and average scores.

4.2. Implementation Details

We pretrain LLaVA-Prime-7B and LLaVA-Prime-72B with data of LLaVA-One-Vision [38] and introduce Panda-70M [5] as the video-text dataset. We pretrain LLaVA-Prime-7B models following the same configuration as LLaVA-One-Vision-7B with an additional 32M video samples from Panda-70M (25B video tokens, 1 fps). For

Method	zero-shot **text** retrieval						zero-shot **image** retrieval						
	Flickr30K			COCO			Flickr30K			COCO			
	R@1	R@5	R@10	R@1	R@5	R@10	R@1	R@5	R@10	R@1	R@5	R@10	MR
EVA-01-CLIP-g/14	87.9	98.0	99.5	61.7	83.2	89.9	72.5	91.5	95.4	44.5	69.1	77.7	80.9
EVA-01-CLIP-g/14+	93.3	99.5	99.9	69.4	88.3	93.2	79.2	95.2	97.3	51.1	74.7	82.5	85.3
OpenCLIP-G/14	93.5	99.3	99.7	69.0	87.8	93.1	80.9	95.1	97.2	52.6	76.1	83.6	85.7
EVA-02-CLIP-E/14+	94.3	99.6	99.8	69.4	88.6	93.3	79.7	94.9	97.3	52.5	75.9	83.4	85.7
EVA-02-CLIP-8B	95.6	99.6	99.9	70.3	89.3	93.9	80.8	95.5	97.6	53.0	76.0	83.4	86.2
DFN5B-CLIP-H/14	92.9	99.3	99.9	72.3	90.2	94.6	80.1	95.2	97.3	53.9	78.0	85.6	86.6
InternVL-C-6B	93.8	**99.7**	**100.0**	70.3	89.2	93.8	82.1	96.0	98.1	54.1	77.1	84.8	86.6
DFN5B-CLIP-H/14 +	93.6	99.3	99.6	71.8	90.4	94.9	82.1	96.0	97.9	55.6	**79.2**	**86.3**	87.2
EVA-02-CLIP-18B	**96.7**	**99.7**	**100.0**	**73.6**	**90.9**	**95.0**	**83.3**	**96.3**	**98.3**	**56.2**	78.5	85.6	**87.8**
LLaVA-Prime-ViT-g	93.6	99.7	100.0	70.5	88.9	93.8	81.1	95.6	97.4	53.9	77.1	84.8	86.4

Table 6. **Zero-shot retrieval performance on Flickr30K [76] and COCO [43]**. We report the R@1, R@5, R@10 accuracy scores. **LLaVA-Prime-ViT-g outperforms EVA-02-CLIP-8B**.

LLaVA-Prime-72B, we use an additional 128M video samples (100B video tokens, 1 fps) from Panda-70M with LLaVA-One-Vision dataset. We use 512 NVIDIA A100 GPUs to pretrain LLaVA-Prime-72B models for two weeks, which is about 40000 steps (335B tokens in total) with an initial learning rate of 5×10^{-5} decay to 5×10^{-7}.

4.3. Vision Encoder Evaluation

Zero-shot Image Classification. In Table 4, LLaVA-Prime-ViT-g demonstrates the efficacy of integrating video pretraining into image classification models, achieving a notable average top-1 accuracy of 78.5% across 27 diverse datasets. This model builds on the EVA-01-CLIP-g/14 [18], which attains a comparatively lower average accuracy of 72.4%. Remarkably, LLaVA-Prime-ViT-g achieves these gains with a relatively modest parameter count of 1 billion, demonstrating that video-enhanced pretraining can substantially improve model performance without the computational overhead associated with larger parameter models, such as InternVL-C-6B with 6 billion parameters. LLaVA-Prime-ViT-g's performance highlights the potential of video pretraining as a scalable, parameter-efficient strategy for enhancing zero-shot image classification.

Zero-shot Video Recognition. Table 5 illustrates the performance of zero-shot video classification across Kinetics datasets (K400, K600, and K700), where models are evaluated on top-1 accuracy and average scores. LLaVA-Prime-ViT-g, under 8 frames per video for evaluation, achieves a solid average score of 73.5% across the datasets (73.5% on K400, 71.3% on K600, and 62.0% on K700), placing it among the top-performing models. While larger models, such as VideoPrism-1B, attain slightly higher scores (85.4% average on K400), LLaVA-Prime-ViT-g demonstrates a remarkable balance of efficiency and performance, requiring fewer frames and maintaining competitive accuracy. As evidenced by its strong results in Table 4 for zero-shot image classification, where it achieves an average top-1 accuracy of 78.5% across 26 diverse datasets. This dual strength in both video and image classification is particularly noteworthy, as LLaVA-Prime-ViT-g effectively bridges the gap between modalities with a single model, showcasing impressive generalization across distinct visual tasks.

Zero-shot Retrieval. Table 6 presents the zero-shot retrieval performance of various models on the Flickr30K and COCO datasets for both text and image retrieval tasks. In zero-shot text retrieval, LLaVA-Prime-ViT-g achieves high performance on Flickr30K, with R@1, R@5, and R@10 scores of 93.6%, 99.7%, and 100.0%, respectively. On the COCO dataset, LLaVA-Prime-ViT-g maintains competitive results with 70.5% at R@1, 88.9% at R@5, and 93.8% at R@10. These scores demonstrate LLaVA-Prime-ViT-g's robust retrieval capabilities across general and challenging text retrieval benchmarks. For zero-shot image retrieval, LLaVA-Prime-ViT-g again performs competitively. It achieves an R@1 score of 81.1% on Flickr30K and 53.9% on COCO, closely rivaling other high-performing models like EVA-02-CLIP-18B, which holds the highest R@1 on both datasets.

4.4. Scaling Vision-Language Models with Video

Scaling Settings. We scale the video data of the video pretraining process from 2M to 16 M, 32M, 64M, and 128M pairs (100B video tokens).

Scaling Video for Image Understanding. As shown in Figure 3, scaling pretraining video data significantly boosts performance across vision-language tasks, underscoring the importance of data scale. For general vision-language tasks (MME, MMBench, POPE, SeedBench, MMMU), scores improve from 66.2% at 2M to 72.2% at 128M. In visual question answering (GQA, OKVQA, Wizwiz), perfor-

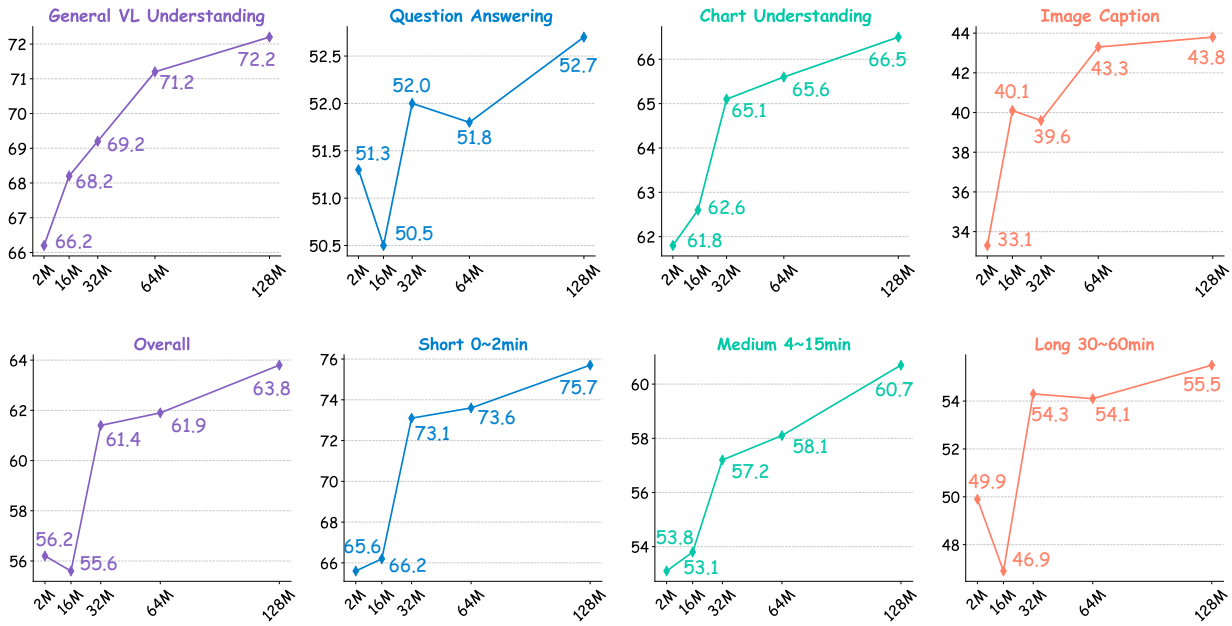


Figure 3. **Scaling Video Data on Image & Video understanding with LLaVA-Prime-7B**. As we scale pretraining video data from 2M to 16M, 32M, 64M, and 128M (about 100B video tokens), we observe significant improvements in both image understanding, including general vision-language understanding (MME, MMBench, POPE, SeedBench, MMMU), question answering (GQA, OKVQA, Wizwiz), chart understanding (TextVQA, DocVQA, InfoQA), image caption (COCO Caption, Nocaps, TextCaps) and video understanding on the VideoMME benchmark including short, medium, and long video understanding.

Method	LLM	Visual Question Answering						Image Caption		MM Benchmark			
		GQA	VQAv2	OKVQA	TVQA	SQA	Vizwiz	NoCaps	Flickr	MME	MMB	MMVet	SEED
Vision Specialist LLM													
BLIP-2 [40]	Vicuna-13B	41.0	41.0	-	42.5	61	19.6	103.9	71.6	1293.8	-	22.4	-
InstructBLIP [12]	Vicuna-13B	49.5	-	-	50.7	63.1	34.3	121.9	82.8	1212.8	-	25.6	-
IDEFICS-80B [35]	LLaMA-65B	45.2	60.0	45.2	30.9	-	36.0	-	53.7	-	54.5	-	-
Qwen-VL [2]	Qwen-7B	57.5	78.2	56.6	61.5	68.2	38.9	120.2	81.0	1487.5	60.6	-	58.2
LLaVA-v1.5 [44]	Vicuna-7B	62.0	78.5	-	58.2	66.8	50.0	-	-	1510.7	64.3	30.5	58.6
LLaVA-v1.5 [45]	Vicuna-13B	63.3	80.0	-	61.3	71.6	53.6	-	-	1531.3	67.7	35.4	61.6
LLaVA-v1.6 [44]	Vicuna-7B	64.2	81.8	-	64.9	70.1	57.6	-	- -	1519	67.4	43.9	70.2
Multimodal Generalist LLM													
AnyMAL-13B [53]	LLaMA2-13B	-	59.6	33.1	24.7	52.7	24.4	-	-	-	-	-	-
AnyMAL-70B [53]	LLaMA2-70B	-	64.2	42.6	32.9	70.8	33.8	-	-	-	-	-	-
OneLLM-7B [26] [CVPR'24]	LLaMA2-7B	59.5	71.6	**58.9**	34.0	63.4	45.9	115.9	78.6	1392.0	60.0	29.1	61.2
LLaVA-OneVision-7B [38]	Qwen2-7B	-	-	-	-	-	-	-	-	1580	80.8	57.5	75.4
LLaVA-OneVision-72B [38]	Qwen2-7B	-	-	-	-	-	-	-	-	2261	85.9	63.7	78.0
LLaVA-Prime-7B	Qwen2-7B	**66.5**	**79.0**	57.8	**67.1**	**80.1**	**58.5**	**122.2**	**84.3**	**1664**	**81.2**	**58.3**	**76.3**
LLaVA-Prime-72B	Qwen2-72B	**74.3**	**83.1**	**68.9**	**72.3**	**84.3**	**67.3**	**135.2**	**88.4**	**2284**	**86.4**	**65.2**	**78.4**

Table 7. **Evaluation on LLM Benchmarks.** The MLLM evaluation involves 6 VQA tasks (GQA [31], VQAv2 [24], OKVQA [52], TextVQA (TVQA) [62], ScienceQA (SQA) [48] and Vizwiz [25]), 2 image captioning tasks (Nocaps [1] and Flickr30K [58]), and 4 multimodal benchmarks (MME [21], MM Bench (MMB) [47], MMVet [77] and SEED [37]). The evaluation metrics for VQA and captioning tasks are accuracy and CIDEr, respectively.

mance rises to 52.7%, showing benefits despite some variability. Chart and document tasks (TextVQA, DocVQA, InfoQA) improve from 61.8% to 66.5%, and image captioning (COCO Caption, Nocaps, TextCaps) scores grow from 33.1% to 43.8%. These results confirm that scaling video data enhances VLMs' robustness and generalization.

Scaling Video for Video Understanding. Similarly, scaling video data leads to substantial gains in video understanding across various video lengths. For general video understanding, scores rise from 56.2% at 2M to 63.8% at 128M, indicating that larger datasets enable more robust comprehension across diverse content. For short videos, performance improves consistently from 66.2% at 2M to 75.7% at 128M, showing that short-form content benefits significantly from increased data exposure. Medium-length videos exhibit a similar trend, with scores increasing

Model	# Frames	Short Video Benchmark							Long Video Benchmark			
		MSVD	ActivityNet	Video-ChatGPT (Score)					EgoSchema	CinePile	VideoMME	MLVU
		(Acc./Score)	(Acc./Score)	CI	DO	CU	TU	CO	Acc.	Acc.	wo/w-subs	m-avg
Proprietary Models												
Gemini 1.0 Pro	-	-	49.8	-	-	-	-	-	55.7	-	-	-
Gemini 1.0 Ultra	-	-	52.2	-	-	-	-	-	61.5	-	-	-
Gemini 1.5 Pro	-	-	56.7	-	-	-	-	-	63.2	61.4	75.0/81.3	-
GPT-4V	-	-	59.5	4.09	3.88	4.37	3.94	4.02	55.6	58.8	59.9/63.3	-
GPT4-O	-	-	61.9	-	-	-	-	-	72.2	59.7	71.9/77.2	66.2
Open-Source Models												
VideoLLaMA (7B)	8	51.6	12.4	1.96	2.18	2.16	1.82	1.79	38.4	-	-/40.4	47.3
Video-ChatGPT (7B)	8	64.9	35.2	2.50	2.57	2.69	2.16	2.20	-	-	-	-
VideoChat (7B)	8	56.3	26.5	2.23	2.50	2.53	1.94	2.24	-	-	-	29.2
Chat-UniVi (7B)	8	65.0	46.1	2.89	2.91	3.46	2.89	2.81	-	-	-/45.9	-
LLAMA-VID (7B)	1 fps	69.7	47.4	2.96	3.00	3.53	2.46	2.51	38.5	-	-	33.2
Video-LLaVA (7B)	8	70.7	45.3	2.87	2.94	3.44	2.45	2.51	-	-	39.9/41.6	-
VideoChat2 (7B)	16	70.0	49.1	3.02	2.88	3.51	2.66	2.81	54.4	-	-/54.6	47.9
LLaVA-NeXT-Video (7B)	32	-	53.5	3.39	3.29	3.92	2.60	3.12	-	-	-	-
PLLaVA (7B)	16	76.6	56.3	3.21	2.86	3.62	2.33	2.93	-	-	-	-
LLaVA-OneVision (7B)	16	-	56.6	3.49	-	-	-	-	60.1	-	58.2	64.7
LLaVA-Prime-7B	16	**78.3**	**62.5**	**3.60**	2.95	**3.98**	**3.01 3.14**	**3.15**	**62.6**	53.5	**63.8**	**65.1**

Table 8. **Zero-Shot video question-answering with LLMs.** We evaluate models by video-based generative performance benchmark [49]. CI DO CU TU CO indicate the evaluation metric of Correctness of Information, Detail Orientation, Contextual Understanding, Temporal Understanding, and Consistency, respectively.

from 53.1% to 60.7% as the dataset scales, highlighting the model's enhanced understanding of moderately temporal context. For long videos, performance fluctuates, initially dipping to 46.9% at 16M but ultimately reaching 55.5% at 128M.

4.5. Vision-Language Model Evaluation

Text-Image Abilities of LLaVA-Prime. Table 7 demonstrates exceptional performance and balance of LLaVA-Prime across a variety of complex tasks. On visual question answering (VQA) tasks, LLaVA-Prime-7B leads with scores of 66.5% on GQA and 80.1% on SQA, highlighting its capacity to handle intricate visual queries with higher accuracy and contextual understanding than models such as Qwen-VL [65]. LLaVA-Prime-7B further distinguishes itself in image caption on Flickr and an unprecedented 58.3 on the MMVet dataset, indicating its capability to generate precise, contextually rich captions—a dimension where both vision specialists and generalist models struggle to achieve comparable precision or descriptive nuance. Beyond task-specific performance, LLaVA-Prime-7B excels in integrative multimodal benchmarks, achieving high scores across MMBench (81.2%), MME (1664), and SEED (76.3%). This consistency across varied benchmarks underscores LLaVA-Prime-7B's ability to integrate visual and linguistic information, which tend to exhibit limitations in either adaptability or consistent accuracy across domains.

Video-Language Ability of VLM. LLaVA-Prime-7B demonstrates substantial advantages in video-based multimodal tasks, as shown in Table 8, presenting powerful performances across both short and long video benchmarks. In the short video benchmark, LLaVA-Prime-7B achieves an impressive accuracy of 78.3% on MSVD, which significantly outperforms LLAMA-VID and PLLAVA, indicating strong capability in short video interpretation. Additionally, LLaVA-Prime leads in ActivityNet with an accuracy of 62.5%, underscoring its effectiveness in activity recognition tasks. Across various Video-ChatGPT subcategories, LLaVA-Prime-7B attains a score of 3.60, showcasing its versatility in handling diverse conversational and task-oriented queries related to video content. In long video benchmarks, LLaVA-Prime-7B maintains its advancement with 62.6% in EgoSchema, highlighting its ability to process egocentric perspectives and long-term dependencies effectively.

5. Conclusion and Discussion

In conclusion, incorporating video pretraining into Vision-Language Models (VLMs) offers a promising solution to the limitations of traditional image-based approaches. By leveraging large-scale video datasets, LLaVA-Prime reduces dependency on scarce, high-quality image-text data while significantly enhancing the model's ability to capture temporal dynamics essential for video understanding. The experimental results demonstrate that VLMs pretrained on video data outperform their image-pretrained counterparts, highlighting the effectiveness of our approach in achieving richer spatial-temporal representations. This advancement not only broadens the scope of multimodal learning in dynamic contexts but also provides a scalable framework for future developments in VLMs, paving the way for more robust and versatile applications across a variety of domains.

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